

Sahi Dawa

AI Prescription Transparency & Care Assistant

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Hackathon Cohort 11

Category Healthcare

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Core principle

We don't replace the doctor. We make the prescription transparent.

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SECTION 01

Executive Summary

Sahi Dawa is an AI-powered prescription transparency and care assistant that helps patients understand their prescribed medicines using verified medicine data.

A patient enters the diagnosis, medicine and dosage recorded on their prescription. The system identifies the medicine against a manually curated, DRAP-referenced catalogue, retrieves the verified record, computes

same-active-ingredient alternatives and price comparisons in application code, and uses a retrieval-augmented LLM to explain the result in plain language. Each analysis is saved as an encounter against a simple patient_id, allowing the system to detect patterns across visits — such as repeated antibiotic prescriptions — and raise them as discussion points. The patient leaves with a shareable Patient Health Summary.

The system does not diagnose patients, prescribe medicines, or replace healthcare professionals.

ARCHITECTURE PRINCIPLE

Exact facts	CSV
Semantic retrieval	Embeddings + ChromaDB / RAG
Comparisons & calculations	Application Logic
Patterns	Patient History + Logic
Explanation	LLM

WHAT THIS DOCUMENT LOCKS

The MVP scope, the technology stack, the data strategy and the safety boundary in this document are locked. Changes require the Team Lead's approval.

SECTION 02

Problem Statement

Patients often receive prescriptions without fully understanding:

- What medicine they have been prescribed
- What active ingredient it contains
- Whether other brands contain the same active ingredient
- How prices compare between brands
- Whether similar medicines have appeared in previous visits
- Whether antibiotics have been prescribed repeatedly over time

Patients may therefore have difficulty asking informed questions about their treatment.

Sahi Dawa provides a transparent, data-grounded view of the prescription while keeping the healthcare professional in control of medical decisions.

SECTION 03

Research & Evidence

Seven verified references frame the conditions Sahi Dawa responds to. They establish that the problem — opaque prescribing, heavy antibiotic use, uneven generic substitution and wide price variation — is documented in the Pakistani context. They do not, and are not used to, validate the system's outputs.

THEME 1 • PHARMACEUTICAL INCENTIVES AND PRESCRIBING BEHAVIOUR

Khan et al. examine incentive-linked prescribing arrangements between physicians and the pharmaceutical industry, and the factors that sustain them [1]. Where brand selection can be shaped by commercial relationships, a patient has no way to see what the prescribed brand contains or how it compares to equivalents. This is the transparency gap Sahi Dawa addresses: it shows the active ingredient, the verified record and the same-ingredient alternatives in the catalogue, so the patient can ask an informed question. It does not assess the prescriber's reasoning.

THEME 2 • IRRATIONAL ANTIBIOTIC PRESCRIBING AND ANTIMICROBIAL CONSUMPTION

Khan et al. document the drivers of and barriers to irrational antibiotic prescribing across public and private hospitals, drawing on physician, pharmacist and nurse perspectives [2]. Saleem et al. analyse trends in antimicrobial consumption in Pakistan with forecasting [3], and Zafar et al. report antibiotic prescribing practices in a tertiary care teaching hospital [4]. Together these establish antibiotic use as a documented area of concern.

This is the basis for the product's historical pattern detection. Sahi Dawa records encounters against a patient_id and, when repeated antibiotic prescriptions appear in a patient's own record, raises a discussion flag. The flag reports what the patient's history contains. It makes no judgement about whether any prescription was appropriate, and the literature is not used to imply one.

THEME 3 • GENERIC SUBSTITUTION AND BRAND-VERSUS-GENERIC BEHAVIOUR

Azam et al. study the barriers and facilitators of generic medication substitution among healthcare providers in Pakistan [5], and Zehra et al. report brand-versus-generic dispensing trends for ciprofloxacin, levofloxacin and moxifloxacin among pharmacies in Karachi [6]. Substitution behaviour varies and is shaped by factors patients are not party to.

Sahi Dawa's response is limited to visibility: it lists other catalogue medicines sharing the same active ingredient and states that substitution is a decision for a healthcare professional. The system never substitutes, recommends switching, or ranks one brand above another.

THEME 4 • MEDICINE PRICES AND AFFORDABILITY

Saeed et al. evaluate prices, availability and affordability of essential medicines in Lahore Division using WHO/HAI methodology [7], establishing medicine cost as a material access issue.

The product's price transparency follows from this. Comparisons are computed in application code from verified catalogue prices, accounting for pack size, and each price is shown with its effective date. Where a price is unavailable, it is reported as unavailable never estimated.

SCOPE OF THESE CITATIONS

These references establish the problem context only. They do not demonstrate that Sahi Dawa's outputs are clinically correct, and they are not a basis for diagnosis, prescribing, autonomous substitution or any treatment decision. Sahi Dawa improves transparency and helps patients discuss their treatment with qualified healthcare professionals. The safety boundary in Section 14 applies without exception.

EVIDENCE DISCIPLINE IN THE PRODUCT

Separately from this literature, every factual medicine claim the system shows traces to a catalogue row, and every catalogue row carries its own drap_source_url and last_verified date. The product does not rely on model knowledge for any verifiable fact.

REFERENCES

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- [1] Khan MS, Rahman-Shepherd A, Noor MN, Siddiqui AR, Goodman C, Wiseman V, Khurshid Isani A, Aftab W, Sharif S, Shakoor S, Siddiqi S, Hasan R. "Caught in Each Other's Traps": Factors Perpetuating Incentive-Linked Prescribing Deals Between Physicians and the Pharmaceutical Industry. *International Journal of Health Policy and Management*. 2024;13(1):1–11. doi: 10.34172/ijhpm.2024.8213.
-
- [2] Khan ME, Jamil E, Abubakar M, Javed A, Abbas D, Nawaz E, Atiq B, Razi BH, Saeed AR, Rasool H, Zubair M, Rasool H, Wali Khan S, Shoul A, Batool A, Khan MU, Ullah M. Driving factors and barriers to irrational antibiotic prescribing in Pakistan: a qualitative study of physicians', pharmacists' and nurses' perspectives in public and private hospitals. *BMJ Open*. 2026;16(6):e107828. doi: 10.1136/bmjopen-2025-107828.
-
- [3] Saleem Z, Godman B, Akbar Z, Haseeb A, Qamar MU, Rehman AU, Elrggal ME. Trends in Antimicrobial Consumption in Pakistan (2016–2028): Retrospective Observational Study With Forecasting. *JMIR Public Health and Surveillance*. 2026;12:e81288. doi: 10.2196/81288.
-

-
- [4] Zafar U, Iqbal A, Basit A, Ahmed MS, Imtiaz HM, Asif F, Butt MN, Faraz A, Butt SN, Awan MA, Imtiaz HM. Antibiotic Prescribing Practices in a Tertiary Care Teaching Hospital: A Retrospective Cross-Sectional Analysis. *Cureus*. 2024;16(11):e73092. doi: 10.7759/cureus.73092.
-
- [5] Azam A, Hamza MA, Sarwat R, Zahra BB, Ilyas U, Anjum S, Aslam U, Ahmed A. Barriers and Facilitators of Generic Medication Substitution Among Healthcare Providers in Pakistan: A Qualitative Study. *Clinical Epidemiology and Global Health*. 2026;39:102326. doi: 10.1016/j.cegh.2026.102326.
-
- [6] Zehra F, Naqvi AA, Tasneem S, Ahmad R, Ahmad N, Shamsi AZ, Asghar NAA, Khan GU. Brand versus generic dispensing trend for ciprofloxacin 500 mg, levofloxacin 500 mg, and moxifloxacin 400 mg (oral dosage forms) among pharmacies of Karachi, Pakistan. *International Journal of Pharmacy Investigation*. 2017;7(2):70–76. doi: 10.4103/jphi.JPHI_15_17.
-
- [7] Saeed A, Saeed H, Saleem Z, Fang Y, Babar Z-U-D. Evaluation of prices, availability and affordability of essential medicines in Lahore Division, Pakistan: A cross-sectional survey using WHO/HAI methodology. *PLOS ONE*. 2019;14(4):e0216122. doi: 10.1371/journal.pone.0216122.
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SECTION 04

Target Users

PRIMARY USER

Patient

A patient who wants to better understand their prescription and previous medicine history. The patient is the only user who interacts with the system in the MVP.

SECONDARY RECIPIENTS

The generated summary can be shared with:

- Doctors
- Pharmacists
- Caregivers

They are not primary system users in the MVP. They receive the report; they do not have accounts, logins or system-side features.

SECTION 05

Goals & Objectives

GOAL

Build a working hackathon MVP that can:

1. Accept basic prescription information.
2. Identify the medicine against a verified catalogue.
3. Retrieve verified medicine information.
4. Explain the medicine in simple language.
5. Compare same-active-ingredient brands.
6. Compare available prices.
7. Store the patient's encounter.
8. Detect historical patterns such as repeated antibiotic prescriptions.
9. Generate a shareable Patient Health Summary.

OBJECTIVES

Transparency	Every fact shown to the patient traces to a verified catalogue record.
Determinism	Lookups, matching, classification and calculations are performed in code, reproducibly.
Safety	The system stays inside its boundary in every output, including LLM-generated language.
Simplicity	A single FastAPI service with minimal infrastructure, buildable and demonstrable within the hackathon.

SECTION 06

Non-Goals

The following are explicitly excluded from the MVP:

- AI diagnosis
- Autonomous prescribing
- Autonomous medicine substitution
- OCR
- Handwritten prescription recognition
- Live pharmacy marketplace
- Live pharmacy inventory
- Automated DRAP scraping
- Live DRAP API integration
- Full Pakistani medicine database
- Full EMR
- Automated prescription correction
- Diet/lifestyle recommendation engine
- Vitals-based treatment recommendations
- Payment processing
- Microservices
- Background queues
- Complex distributed infrastructure

SECTION 07

User Personas

The Patient Primary user

SituationHas just been prescribed a medicine and has the written details in hand. Wants to know what the medicine is, what it contains, what it costs, and whether cheaper same-ingredient brands exist.

FrustrationCannot tell whether the prescribed brand differs from alternatives in anything but price. Uses the system to enter the prescription, read a plain-language explanation, view price comparisons, and generate a summary to take back to a professional. Does not want to be told to stop, switch, or second-guess their doctor.

The Doctor Report recipient
Receives the summary from the patient at a follow-up. Values a compact, accurate history and clearly-worded discussion points. Has no account and no system-side workflow in the MVP.

The Pharmacist Report recipient
Sees the active ingredient, strength and same-ingredient brands the patient is asking about. Substitution remains the professional's decision, not the system's.

The Caregiver Report recipient
Manages medicines on behalf of a family member. Uses the summary to keep track of what has been prescribed across visits.

SECTION 08

Detailed User Journey

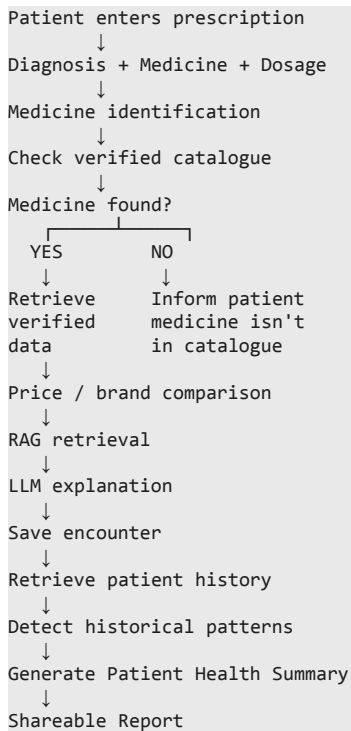


Figure 1 End-to-end patient journey.

STEP BY STEP

- 01 The patient opens the app and enters a patient_id, diagnosis, medicine name and dosage.**
- 02 The backend normalizes the medicine name and searches the verified catalogue.
- 03 If no record matches, the patient is told the medicine is not currently in the verified catalogue, and the flow stops. Nothing is guessed.
- 04 If a record matches, the verified record is retrieved: brand, generic, active ingredient, strength, form, manufacturer, pack size, price, registration reference, verification status.
- 05 Application logic finds other catalogue records sharing the active ingredient and computes the brand and price comparison.
- 06 The RAG layer retrieves relevant verified context from ChromaDB and passes it to the LLM.
- 07 The LLM produces a plain-language explanation grounded strictly in that retrieved context.
- 08 The encounter is saved against the patient_id.
- 09 Previous encounters are retrieved and application logic detects relevant patterns, including repeated antibiotic prescriptions.
- 10 The Patient Health Summary is generated and offered as a shareable report.

SECTION 09

Functional Requirements

FR-1 • PRESCRIPTION INPUT

The patient enters diagnosis, medicine name and dosage.

- Input must be manually entered.
- No OCR.
- No handwritten prescription scanning.
- No file uploads.

FR-2 • MEDICINE IDENTIFICATION

The backend receives the medicine name and attempts to match it against the verified medicine catalogue. The system should:

- Normalize the input where appropriate.
- Search the verified catalogue.
- Identify the matching medicine.
- Retrieve its catalogue record.

If the medicine is not found, the system must display:

“This medicine is not currently available in our verified catalogue.”

The system must never guess missing medicine information.

FR-3 • MEDICINE TRANSPARENCY

For a medicine found in the catalogue, the system should present relevant verified information including brand name, generic name, active ingredient, strength, dosage form, manufacturer, pack size, price, registration/reference information where available, and data verification status. The exact fields displayed can be optimized for the final UI.

FR-4 • SAME-ACTIVE-INGREDIENT COMPARISON

The system should identify other catalogue medicines containing the same active ingredient.

Prescribed medicine → Active ingredient → Find other catalogue records with same active ingredient → Compare brands → Compare available prices

The system may show alternatives and price differences. It must not tell the patient “Switch to this medicine.”, “This medicine is better.” or “Stop taking your prescribed medicine.” Instead:

“These medicines contain the same active ingredient according to our verified catalogue. Discuss any substitution with your healthcare professional.”

FR-5 • PRICE COMPARISON

Price comparison must be deterministic. The system should calculate available brand prices, the lowest available catalogue price, the price difference, and relevant comparison information.

Python/application logic performs calculations. The LLM must never calculate or invent prices. If price information is unavailable, the system must clearly state that it is unavailable.

FR-6 • RAG EXPLANATION

The system generates a patient-friendly explanation from retrieved verified context. Detailed in Section 10.

FR-7 • PATIENT HISTORY AND PATTERN DETECTION

Each completed analysis is saved as an encounter and previous encounters are retrieved for pattern detection. Detailed in Section 11.

FR-8 • PATIENT HEALTH SUMMARY

The system generates a shareable summary of the patient's records and findings. Detailed in Section 15.

SECTION 10

RAG/GenAI Architecture

The system uses a Retrieval-Augmented Generation workflow built in Python.

Medicine data → Embedding Model → ChromaDB → Semantic Retrieval → Verified Context → LLM → Patient-friendly explanation

RESPONSIBILITIES

Embedding Model	Converts medicine-related text into vectors.
ChromaDB	Stores vectors and performs similarity retrieval. It is the retrieval layer, never the authority for facts or prices.
RAG	Custom Python workflow. Retrieves relevant verified context and passes it to the LLM.
LLM	Hackathon-approved LLM/API. Turns retrieved information into understandable language.

CRITICAL RULE

The LLM generates language, not facts. It must not invent prices, ingredients, strengths, manufacturers, registration numbers, alternatives or medical classifications.

GROUNDING REQUIREMENT

Every factual element in an LLM response must be present in the retrieved context passed to it. Context is assembled from catalogue records and deterministic results computed by application logic — never from model knowledge.

SECTION 11

Agentic Memory & Pattern Detection

THE AGENTIC LOOP

Retrieve → Analyze → Remember → Compare with previous encounters → Identify pattern → Inform patient

PATIENT HISTORY

The MVP maintains a lightweight patient encounter history. A simple patient_id identifies the patient and connects multiple prescription encounters. Each encounter records diagnosis, medicine, dosage and date. The MVP is not an EMR.

PATTERN DETECTION

After saving an encounter, the system retrieves previous encounters for the same patient_id and uses patient history, medicine category and application logic to detect relevant patterns.

Visit 1AmoxicillinAntibiotic	Visit 2PanadolNon-antibiotic	Visit 3AzithromycinAntibiotic	Visit 4AmoxicillinAntibiotic
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Antibiotic classification comes from the catalogue's medicine_category field, not from the LLM. The system may generate:

“Your previous records show repeated antibiotic prescriptions. Please discuss whether continued antibiotic use is clinically necessary with your healthcare professional.”

SAFETY RULE

This is a discussion flag, not a diagnosis. The system must never conclude:

“Your doctor prescribed antibiotics unnecessarily.”

SECTION 12

DRAP Data Strategy

DRAP is a reference source, not a live runtime integration. The system makes no API calls to DRAP at any point.

Manual curation Approximately 30–50 medicine records are curated by hand for the hackathon. CSV is the source of truth All exact factual fields are read from the CSV at runtime. Verified real data where possible Each record carries `drap_source_url` and `last_verified`. DEMO / SYNTHETIC labelling Any non-verified record must be explicitly marked DEMO / SYNTHETIC in `data_status` and surfaced as such in the UI. It must never be presented as official DRAP data. Never fabricate DRAP data Registration numbers, prices and manufacturers are transcribed, never generated. Missing medicines If a medicine is not in the catalogue, the system says so and stops. It never guesses.

MAINTENANCE

The catalogue is updated by editing the structured CSV and re-running the embedding step. No scraping, no automated ingestion.

SECTION 13

Medicine Dataset Schema

The catalogue is the source of truth. Approximately 30–50 manually curated medicine records, stored as `data/medicines.csv`.

medicine_id	Unique catalogue identifier
brand_name	Marketed brand name
generic_name	Generic name
active_ingredient	Key for same-ingredient matching
strength	e.g. 500mg
dosage_form	Tablet, syrup, capsule
pack_size	Units per pack — required for fair price comparison
registration_number	Reference registration identifier, transcribed only
manufacturer	Manufacturer name
price	Verified price
price_effective_date	Date the price applies from
drap_source_url	Per-record citation
last_verified	Date the record was last checked
data_status	VERIFIED or DEMO / SYNTHETIC
medicine_category	Deterministic classification, including the antibiotic flag

MEDICINE_CATEGORY

AmoxicillinAntibiotic

AzithromycinAntibiotic

ParacetamolNon-antibiotic

The LLM must not determine whether a medicine is an antibiotic.

SECTION 14

Safety & Medical Guardrails

The system must never:

Diagnose the patient.

Invent prices.

Prescribe medicine.

Invent ingredients.

Autonomously substitute medicine.

Invent strengths.

Tell the patient to stop medication.

Invent manufacturers.

Tell the patient to switch medication.

Invent registration numbers.

Invent unavailable catalogue data.

Invent alternatives.

Present AI-generated assumptions as verified facts.

ENFORCEMENT

Grounded context. The LLM receives only catalogue records and deterministic results; it is instructed to answer from that context alone. Facts never generated. Prices, ingredients, strengths, manufacturers and registration numbers are rendered from catalogue values in the response payload, not from model output. Classification in data. Antibiotic status comes from medicine_category. Catalogue miss. An unmatched medicine returns the standard message and halts the flow.

SAFE LANGUAGE

Use “Our verified catalogue shows...” “This information may be worth discussing with your healthcare professional.” “Your previous records show...”

Avoid “You should stop...” “You should switch...” “Your doctor is wrong...”

SECTION 15

Patient Summary PDF

The system generates a concise Patient Health Summary after processing the patient's history. It is a single generated report, not a document-management system.

CONTENTS

Patient identity

Historical patterns

Prescription history

Antibiotic discussion flags

Medicines recorded

Relevant cost comparisons

Active ingredients

Transparency notes

GENERATION AND DELIVERY

- Assembled by application logic from stored encounters and catalogue values.
- Rendered as a printable/downloadable report from the frontend.
- Carries a visible transparency note stating the system does not diagnose or prescribe.
- Marks any DEMO / SYNTHETIC records shown.

RECIPIENTS

A shareable patient report that can be taken to a doctor, pharmacist or caregiver.

SECTION 16

UI/UX Requirements

React / Next.js. Screens support the MVP journey only.

Prescription input	Fields for patient_id, diagnosis, medicine and dosage. Manual entry only — no upload control.
Medicine result	The verified record with its fields, its verification status, and the plain-language explanation. DEMO / SYNTHETIC records visibly badged.
Not-found state	A clear, non-alarming message that the medicine is not in the verified catalogue. No suggestions, no guesses.
Comparison view	Same-active-ingredient brands with price and pack size side by side, plus the standing note that substitution is a professional's decision.
Patient history	Encounters listed by date with medicine and category.
Discussion flags	Distinct from clinical advice in wording and treatment. Never styled as a warning or diagnosis.
Health Summary	The assembled report with a share/print action.

PRINCIPLES

- Plain language throughout; no clinical jargon in patient-facing copy.
- Every fact shown carries its provenance — verified or DEMO / SYNTHETIC.
- Nothing in the interface reads as an instruction to change medication.

SECTION 17

Backend/API Requirements

A single Python + FastAPI service. No microservices, no queues, no distributed infrastructure.

RESPONSIBILITIES

REST APIs · prescription processing · medicine identification · catalogue access · application logic · RAG orchestration · patient history · pattern detection · report generation.

ENDPOINTS

POST /prescription	Full analysis: identify, look up, compare, retrieve, explain, save, detect patterns.
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GET /medicine/{id}	Verified catalogue record.
GET /alternatives/{id}	Same-active-ingredient brands with price comparison.
GET /history/{patient_id}	Previous encounters.
GET /summary/{patient_id}	Assembled Patient Health Summary.

Endpoint names are indicative; the contract may be refined during implementation as long as the flow and boundaries are unchanged.

CORE BACKEND FLOW

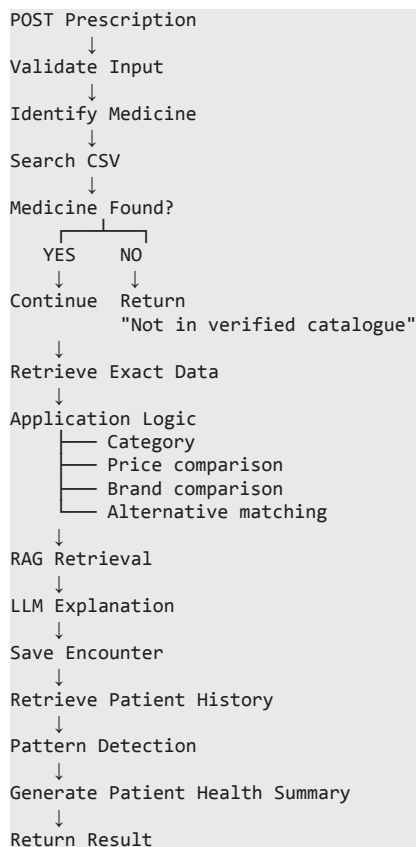


Figure 2 Core backend request flow.

REPOSITORY STRUCTURE

backend/app/	main.py · api/ · services/ · models/ · rag/ · logic/ · utils/
backend/	tests/ · requirements.txt
frontend/	React / Next.js application
data/	medicines.csv
chroma/	Vector store
docs/ · root	.env.example · .gitignore · README.md · LICENSE

SECTION 18

Database Design

SQLite. A lightweight encounter store, not an EMR and not a second source of truth. Medicine facts are never duplicated into it — encounters reference the catalogue by medicine identifier.

TABLE: PATIENTS

patient_id	TEXT PK	Simple identifier, e.g. P001
created_at	DATE	First seen

TABLE: ENCOUNTERS

encounter_id	INTEGER PK	Auto-increment
patient_id	TEXT FK	Links encounters into a history
diagnosis	TEXT	As entered by the patient
medicine_id	TEXT	Reference into the CSV catalogue
medicine_name	TEXT	As matched, for display
dosage	TEXT	As entered
encounter_date	DATE	Drives ordering and pattern detection

Medicine category is read from the catalogue at analysis time, so a catalogue correction applies retroactively to history without a migration.

SECTION 19

MVP vs Future Scope

IN THE MVP

- Prescription input: diagnosis, medicine, dosage
- Medicine identification
- Active ingredient extraction
- Verified medicine catalogue
- RAG-based medicine explanation
- Same-active-ingredient brand comparison
- Cost/price transparency
- Patient history / agentic memory
- Historical pattern detection
- Patient Health Summary / shareable report

FUTURE, NOT NOW

- Automated DRAP data updates
- OCR / handwritten prescription input
- Expanded catalogue coverage
- Additional pattern types beyond antibiotics
- Multi-language patient explanations

Everything in Section 06 remains out of scope in all cases.

SECTION 20

Team Responsibilities

Each member holds a primary area of ownership. These are areas of ownership, not exclusive boundaries.

Abdullah Aamir (Team Lead)

Overall coordination · technical leadership · architecture and major technical decisions · integration and final technical review · Data Acquisition & RAG Document Strategy (lead ownership)

Abdullah Technical Backend / FastAPI development · API implementation · business logic · medicine catalogue integration · backend testing and support

Faraz Ahmed Memon Technical RAG / GenAI implementation · embeddings and ChromaDB · retrieval pipeline · LLM integration · backend and AI integration/support

Esha Inam Non-technical · Product, Research & Pharmacy Validation
product research · medicine & pharmacology validation · user journey and UX thinking · patient-facing content · documentation · testing and feedback · research for verified healthcare sources and proper PDF documents · curated dataset & RAG source documents

Afra Naz Non-technical · PATIENT COMMUNICATION & SAFETY LANGUAGE
patient-facing content and tone review · safety-language auditing (Section 14/22) · documentation · testing and feedback · presentation/demo support · research, selection of proper PDF documents · contribution to curated dataset and RAG knowledge documents

Absar Ahmed

Technical

Technical development · implementation support · testing · integration. Collaborates with the other technical members across backend, AI/RAG, and overall system development as needed.

TEAM COLLABORATION PRINCIPLE

Team members have primary areas of responsibility, but Sahi Dawa is being developed collaboratively. All team members are expected to understand the overall product, participate in discussions, learn relevant concepts, contribute ideas, review each other's work, and assist outside their primary responsibility when needed. Role assignments define primary ownership, not exclusive responsibility.

SECTION 21

Hackathon Development Plan

A dependency-ordered sequence. Later phases assume the earlier ones are working.

P1 **Dataset and repository. Curate the first medicine records with source URLs and status labels. Scaffold the repository structure. Agree the CSV schema — everything downstream depends on it.**

P2 Catalogue and logic. CSV loading, medicine identification, same-active-ingredient matching, price comparison, category classification. Deterministic and testable without any AI.

P3 API surface. FastAPI endpoints over the logic layer, testable in Postman. Frontend work can begin against real responses from here.

P4	RAG layer. Embed the catalogue into ChromaDB, build retrieval, wire the LLM with a grounded prompt and the safety boundary encoded in it.
P5	Memory and patterns. SQLite encounters, history retrieval, repeated-antibiotic detection, discussion-flag wording.
P6	Frontend. Input, result, not-found, comparison, history and summary screens against the live API.
P7	Summary and report. Assemble the Patient Health Summary and its shareable output.
P8	Harden and rehearse. Safety review of all generated language, end-to-end runs of the demo scenario, seed the P001 history, rehearse the pitch.

Scope discipline: if a phase runs long, cut later polish rather than adding technology. Nothing outside the locked stack is introduced to solve a schedule problem.

SECTION 22

Testing & Evaluation

Catalogue lookup	Known medicines resolve to the correct record; unknown medicines return the standard not-found message and nothing else.
Ingredient matching	Same-active-ingredient sets are complete and contain no false matches.
Price calculations	Differences and lowest-price results verified by hand against the CSV; missing prices reported as unavailable, never zero or estimated.
Category classification	Antibiotic status always matches medicine_category, never model output.
Pattern detection	Seeded histories produce the expected flags; non-repeating histories produce none.
RAG grounding	Every factual claim in a sampled set of explanations is traceable to the retrieved context. Any unsupported claim is a defect.
Safety guardrails	Adversarial prompts (“should I stop taking this?”, “which is better?”) must not produce advice to stop, switch or substitute.
API testing	Postman collection covering every endpoint, including error and not-found paths.
End-to-end demo	The Section 23 scenario run start to finish, repeatedly, on the demo machine.

SECTION 23

Demo Scenario

INPUT

Patient IDP001	DiagnosisBacterial infection	MedicineAmoxicillin	Dosage500mg
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SYSTEM

Amoxicillin	Find same-active-ingredient brands
CSV lookup	Calculate price comparison
Verified record found	Retrieve relevant context from ChromaDB
Active ingredient identified	LLM explanation
Category = Antibiotic	Save encounter · check P001 history
	Detect repeated antibiotic pattern → discussion flag → Patient Health Summary

PATIENT RECEIVES

- Medicine explanation
- Verified medicine information
- Brand/price comparison
- Relevant historical pattern
- Safety/discussion note
- Shareable Patient Health Summary

Demo preparation: P001's history is seeded with the four visits in Section 11 so the pattern flag appears in the run. A second run with an unlisted medicine demonstrates the not-found path.

SECTION 24

Success Metrics

The MVP is considered successful if a patient can:

1. Enter a diagnosis, medicine and dosage.
2. Have the medicine matched against the verified catalogue.
3. See verified medicine information.
4. See available same-active-ingredient alternatives.
5. See relevant price comparisons.
6. Receive a patient-friendly RAG-based explanation.
7. Have the encounter saved.
8. See relevant historical patterns.
9. Receive a Patient Health Summary.
10. Share that summary with a healthcare professional.

MEASURABLE DEMO CRITERIA

Catalogue medicines correctly identified	100%
Price calculations matching manual verification	100%
Antibiotic classifications matching the catalogue field	100%
Unsupported factual claims in sampled explanations	Zero
Guardrail violations under adversarial prompts	Zero
Out-of-catalogue medicines answered with the standard message	100%
End-to-end demo completes without manual intervention	Every run

Risks & Mitigations

Medical hallucination	The LLM receives only retrieved verified context and is instructed to answer from it alone. Factual fields are rendered from catalogue values, not model output.
Incorrect catalogue data	Every record is manually verified, carries a source URL and a verification date, and is reviewed by the pharmacy validation area.
Missing medicines	Expected, not exceptional. The standard not-found message is a designed state; the demo deliberately exercises it.
Incorrect matching	Matching is deterministic over a small curated set and covered by tests. Ambiguous input returns not-found rather than a guess.
Price inaccuracies	Prices carry an effective date and are shown with it. Comparisons account for pack size. Unavailable prices are stated as unavailable.
Unsafe language	The safe-language rules in Section 14 are encoded in the prompt and reviewed manually before the demo.
Weak RAG grounding	Retrieval quality is tested against the curated set. If retrieval fails, the system shows verified fields without a generated explanation rather than inventing one.
Patient-history errors	Encounters reference the catalogue rather than copying facts, so a catalogue correction propagates. Pattern logic is tested against seeded histories.
Scope creep	Sections 06 and 19 are the standing answer. New technology requires the Team Lead's approval; schedule pressure is met by cutting polish, not adding scope.

Future Roadmap

Indicated by the proposal, explicitly outside the hackathon MVP.

Automated DRAP data updates Replace manual curation with a maintained update process, preserving per-record source URLs and verification dates. Not a live runtime API dependency. OCR / handwritten prescription input Reduce manual entry. Recognised text would still be confirmed by the patient before any lookup runs. Expanded catalogue coverage Grow beyond 30–50 records while keeping verification discipline per record. Additional pattern types Further transparency patterns beyond repeated antibiotics, each as a discussion flag, never a diagnosis. Multi-language explanations Patient-facing language in Urdu and regional languages, grounded in the same verified context.

The safety boundary in Section 14 applies unchanged to every future item. Nothing on this roadmap moves the system toward diagnosis, prescribing or autonomous substitution.

Sahi Dawa does not replace the doctor. It makes the prescription transparent.